

Experiment No. 7: Customer Segmentation using K-Means Clustering

Aim

To implement the K-Means Clustering algorithm on a real-world dataset for customer segmentation and analyze customer groups based on purchasing behavior and demographic characteristics.

Questions & Answers

Q1. What is clustering? How does it differ from classification?

Answer:

- **Clustering:** An unsupervised learning task that partitions unlabeled data points into distinct groups (clusters) based on intrinsic pattern similarities without predefined ground-truth labels.
- **Classification:** A supervised learning task that trains models on labeled historical datasets to map features to target category classes.

Aspect	Clustering	Classification
Learning Type	Unsupervised Machine Learning	Supervised Machine Learning
Data Labeling	Unlabeled data	Pre-labeled target classes
Goal	Group data points by natural affinity	Predict predefined class categories

Q2. Explain the working principle of the K-Means Clustering algorithm.

Answer:

K-Means partitions N observations into K distinct clusters through an iterative optimization procedure:

1. **Initialization:** Select K initial centroid coordinates randomly across feature space.
2. **Assignment Phase:** Calculate Euclidean distance between every data point and each centroid; assign points to their closest centroid.
3. **Update Phase:** Recalculate cluster centroids as the arithmetic mean of all data points currently belonging to that cluster.
4. **Convergence:** Repeat the assignment and update steps until centroid movements fall below a predefined tolerance threshold or cluster memberships stabilize.

Q3. Why is feature scaling important before applying K-Means clustering?

Answer:

K-Means utilizes distance calculations (typically **Euclidean distance**) to compute similarity between observations.

- Unscaled features measured on larger numeric scales (e.g., Annual Income: ₹15,000 - ₹140,000) will mathematically dominate distance equations compared to features on smaller numeric scales (e.g., Age: 18 - 70).
- Feature scaling (via **StandardScaler** or **MinMaxScaler**) transforms features to equivalent scales, ensuring equal feature weighting.

Q4. What is the Elbow Method, and how does it help determine the optimal number of clusters?

Answer:

- **Elbow Method:** A heuristic optimization tool used to locate the ideal number of clusters (K).
- **Mechanism:** Plots the **Within-Cluster Sum of Squares (WCSS)** / Inertia against a range of sequential K values.
- **Selection Criterion:** As K increases, WCSS naturally drops. The optimal K value corresponds to the visual "elbow point"—the point after which diminishing returns occur in WCSS reduction.

Q5. What is the Silhouette Score? How is it used to evaluate clustering performance?

Answer:

The **Silhouette Score** measures how similar an object is to its own cluster compared to neighboring clusters.

• **Formula:**

$$S(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))}$$

Where:

- $a(i)$ = Mean intra-cluster distance for point i.
- $b(i)$ = Mean nearest-cluster distance for point i.
- **Range Interpretation:**
 - Score range: **-1 to +1**.
 - Near +1: Points are tightly packed within their cluster and far from neighboring clusters (Optimal).
 - Near 0: Overlapping clusters.
 - Negative value: Incorrect cluster assignments.

Q6. Why is K-Means considered an unsupervised machine learning algorithm?

Answer:

K-Means operates entirely on unlabeled training datasets (X). It does not rely on target response variables (y) or predefined guidance labels to group observations. Instead, it uncovers natural structure and hidden groupings based purely on statistical distances.

Q7. Mention any four real-world applications of customer segmentation.

Answer:

1. **Targeted Marketing Campaigns:** Tailoring promotional emails and advertisements to specific demographic clusters.
2. **Personalized Product Recommendations:** Recommending items based on historical purchase patterns of similar customer groups.
3. **Dynamic Pricing Strategy:** Offering targeted discounts to price-sensitive customer segments while providing premium tiers to high-value cohorts.
4. **Customer Churn Prevention:** Identifying declining engagement patterns in specific segments to trigger retention campaigns.

Q8. What are the limitations of the K-Means algorithm?

Answer:

1. **Sensitivity to Initial Centroids:** Random initial centroid selection can yield sub-optimal local minima solutions (mitigated by K-Means++).
2. **Predefined K Value:** Requires manual definition of cluster counts before running execution loops.
3. **Outlier Sensitivity:** Outlier data points pull and distort average centroid positions.
4. **Spherical Cluster Assumption:** Struggles with complex non-convex, non-spherical spatial cluster geometries.

Q9. How can businesses use customer segmentation to improve marketing and customer retention?

Answer:

- **Tailored Messaging:** Replaces generic mass broadcasts with cohort-specific advertising copy that addresses unique buyer preferences.
- **Budget Optimization:** Allocates high marketing budgets toward high-value, high-margin customer cohorts.
- **Proactive Loyalty Programs:** Identifies vulnerable high-spending clusters nearing inactivity and offers targeted incentives to retain them.

Q10. Compare K-Means Clustering with Hierarchical Clustering based on their working principles and applications.

Answer:

Feature	K-Means Clustering	Hierarchical Clustering
Working Principle	Iterative centroid assignment based on Euclidean distances	Builds a nested tree structure (Dendrogram) via Agglomerative or Divisive steps
Speed & Scalability	Fast $O(N \cdot K \cdot I)$, scales well to massive datasets	Computationally heavy $O(N^3)$, restricted to smaller/medium datasets
Cluster Shape	Assumes spherical, convex cluster shapes	Capable of discovering arbitrary, non-spherical cluster forms
Pre-specifying K	Requires defining K in advance	Does not require pre-specifying K; dendrogram can be cut at desired height